INCH-POUND

MIL-C-24643/46D
22 November 1994
SUPERSEDING
MIL-C-24643/46C
14 March 1994

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS2WAU
(INCLUDING VARIATION LS2WA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LS2WAU (unarmored) and LS2WA (armored).

REQUIREMENTS:

Qualification required.

Construction (watertight)

First	- Copper conductor, tin coated, ASTM B 286, size 22-7. maximum diameter 0.033 inch.
Second	- Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	- Crosslinked polyethylene insulation. Nominal wall thickness 0.024 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Telephone identification applied by method 3.
Fourth	- Two conductors cabled together with a nominal lay of 1-1/2 inches to form a pair.
Fifth	- The specified number of pairs (see Table I) cabled together with a lay not greater than 16 times the overall diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from the center outward. Fillers shall be used as necessary to make a firm, well rounded assembly.
* Sixth *	- Binder tape applied, at manufacturer's option, helically with overlap.
Seventh	- Braided shield of tin coated copper. Minimum coverage of 85 percent.
Eighth	- Binder tape, applied helically with overlap.

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Construction (watertight) (continued)

Construction variant for type LS2WAU

Ninth - Cross-linked polyolefin jacket. Minimum average thickness 0.060 inch.

Construction variant for type LS2WA

Ninth - Cross-linked polyolefin jacket. Minimum average thickness 0.060 inch.
Tenth - Braided metal armor.

TABLE I. Details.

Military1/ part no. M24643/46	Size 2/	Number of pairs	Diameter over Shielded core (inch nom)	Overall diameter (inches)			
				TYPE LS2WAU		TYPE LS2WA	
				min	max	min	max
-01UO	40	40	1.090	1.320	1.370	1.370	1.420

1/ Unarmored, overall shield. When armor option is required, see MIL-C-24643 for configuration of part number.
*2/ Type LS2WAU and LS2WA.

EXAMINATION AND TESTS:	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum	17.71
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2500
Insulation resistance - Megohms/1000 feet minimum	
Conductor to conductor	500
Jacket flaws	No failure
Conductor and shield continuity	No failure
Group A:	
Visual and dimensional	No failure
Capacitance	
Mutual capacitance - 1 kilohertz (KHz), pF/ft, maximum	30
Watertightness - See MIL-C-24643 for limits of water leakage	No failure
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (when tested at 200°C)	50
Cold bending, cable - at minus 0 ± 2°c, mandrel diameter 12 times cable diameter	No damage
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35

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EXAMINATION AND TESTS (continued):	Requirements
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Air Oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (jacket) - cycles, minimum	125
Armor (type LS2WA only) - conformance to material, construction and coverage	No failure
Shield - conformance to material, construction and coverage	No failure
Surface transfer impedance	
Milliohms per meter, maximum	700
Emp response time - db, minimum	60
Group D:	
Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Shield Insulation	2

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QUALIFICATION INSPECTION (continued):	Requirements
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS: All sizes 1000 feet (nominal)

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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